

Synchronisation Wand

– An Embedded Hardware Solution to Synchronise Wearables Using Magnetometers

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Background and Project Goal

New Method !

Multiple Inertial Measurement Units are commonly used in many applications. **Synchronisation** is vital.

1. Real Time Clock (RTC) Synchronisation

- ✗ Low Accuracy – Clock Drift

2. Kinetic Event (KE)

- ✗ “Ambiguity” issue

3. Simple Electromagnetic Pulse (Simple EMP)

- ✓ Eliminated “Ambiguity”
- ✗ Limited by low magnetometer T_S

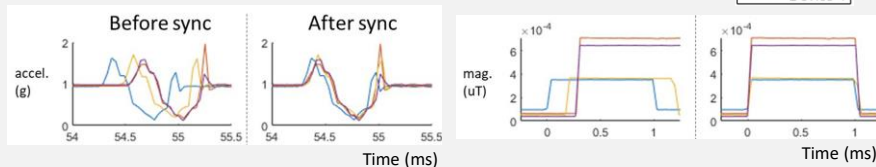


Figure 1. Sync. wearables using KE (left) and simple EMP (right).

4. Encoded Electromagnetic Pulse

- Works by **finding the Extra Sample Pulse**

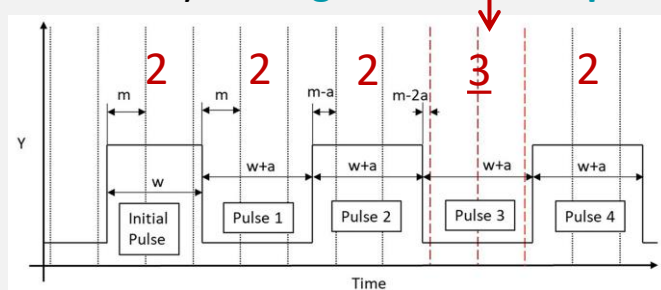


Figure 2. The encoded EMP synchronisation.

- ✓ Eliminated “Ambiguity”
- ✓ 5 ms accuracy with low magnetometer T_S
- ✓ Using existing hardware

Project Goal:

1. **Design a device** for encoded EMP sync.
2. **Fully verify** the device and method.

The Synchronisation Wand

System architecture:

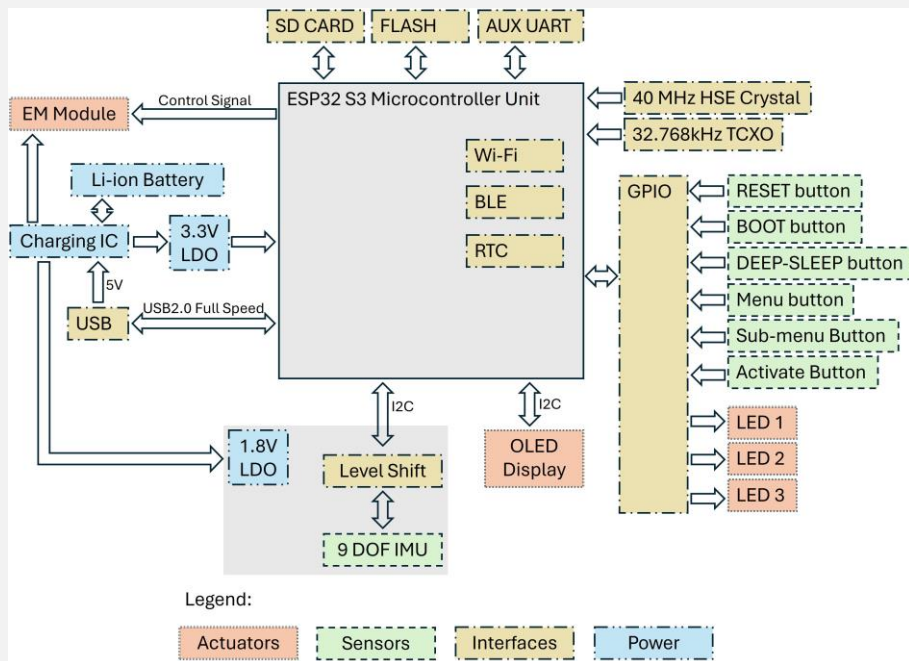


Figure 3. Overview of the Synchronisation Wand architecture.

Firmware Powered by:



Communication Supports:



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PCB Design:

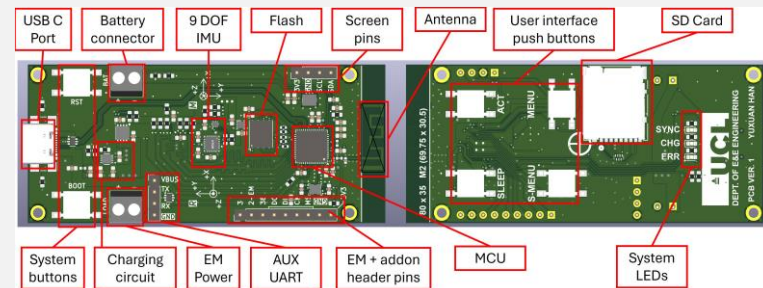


Figure 4. The Synchronisation Wand PCB design.

Verification – FPGA Testbed

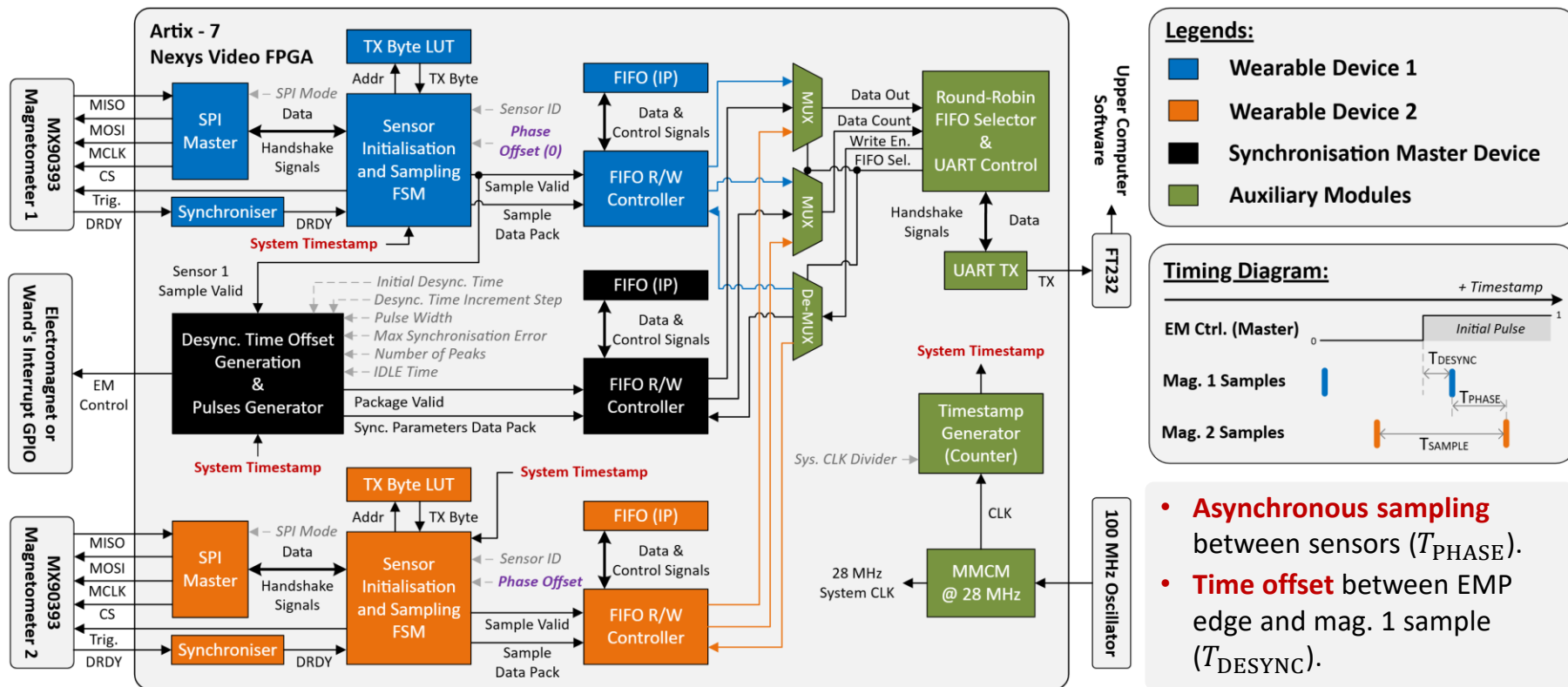


Figure 5. FPGA based testbed structure and sampling timing diagram.

Pg. 5 / 8

Verification Tests and Results

Verification Tests (for 2 scenarios):

1. Sync. time between one wearable and the master device:

- Test pass if *:

$$|p_1 a - T_{\text{DESYNC}}| \leq a$$

(Device 1 → Master) Offset by encoded EMP sync. (points to $p_1 a$)
Real offset (set by testbed) (points to T_{DESYNC})
Max sync. error (points to a)

2. Sync. time between two wearables:

- Test pass if *:

$$|(p_2 - p_1)a - T_{\text{PHASE}}| \leq a$$

(Device 1 → Device 2) Offset by encoded EMP sync. (points to $(p_2 - p_1)a$)
Real offset (set by testbed) (points to T_{PHASE})
Max sync. error (points to a)

Note *: p_n stands for extra sample pulse number. Conditions listed here are for $p_2 \geq p_1$ case only.

Setup:

- 5 test groups:
 $a = 10 \text{ ms}, 5 \text{ ms}, 2 \text{ ms}, 1 \text{ ms}, 0.5 \text{ ms}$
- 400 sync. events per group:
 T_{DESYNC} evenly distributed 0 - 40 ms
- $T_{\text{SAMPLE}} = 40 \text{ ms}, T_{\text{PHASE}} = 2.75 \text{ ms}$

Results:



Sync Accuracy	T1 Pass %	T2 Pass %
0.5 ms	96.03 %	95.04 %
1 ms	100 %	100 %
2 ms	99.5 %	100 %
5 ms	100 %	100 %
10 ms	99.76 %	100 %

Note: Failed events due to imprecise T_{LAG} values.
 All 100% pass if giving an extra 0.1 ms threshold.

Conclusions, Publication & Future Works

Conclusions:

- Successfully made the Synchronisation Wand
- Proposed a new verification method for encoded EMP sync.
- Addressed practical limitations and considerations of the encoded EMP method.

Synchronisation Method	Sync. Error (200 Hz accel. + 25 Hz mag.) ^[1]
KE + Cross-correlation	≤ 5 ms (can be worse due to ambiguity)
Simple EMP	≤ 40 ms
Encoded EMP	≤ 0.5 ms
Encoded EMP (Previous)	≤ 5 ms

Common medical application requirement: ≤ 1 ms ^[2]

[1] M. Karas, J. Bai, M. Strączkiewicz, J. Harezlak, N. W. Glynn, T. Harris, V. Zippunikov, C. Crainiceanu, and J. K. Urbanek, "Accelerometry data in health research: challenges and opportunities," *Statistical Biosciences*, vol. 11, no. 2, pp. 210–237, Jul. 2019, doi: 10.1007/s12561-018-9227-2.

[2] V. Masalsky, D. Čičurėnas, A. Dziedzickis, U. Prentice, G. Braziulis, and V. Bučinskas, "Synchronization of separate sensors' data transferred through a local Wi-Fi network: A use case of human-gait monitoring," *Future Internet*, vol. 16, no. 2, article 36, 2024, doi: 10.3390/fi16020036.

Publication:

- Conference: *ACM Ubicomp / ISWC '24*
- DOI: 10.1145/3675094.3678485
- First author, peer-reviewed.

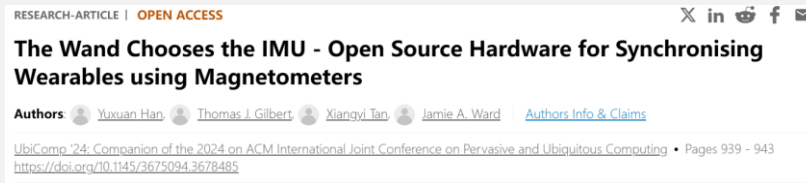


Figure 8. Publication.

Future Work:

- **Further verify** the device under **real-world application** scenarios, e.g. multi-sensor motion capture. (Ongoing)

Thanks !

P.S. This presentation only contains a general overview of my work at a high level, around 30% of all.

Feel free to ask me if you are interested in extra detail.

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